

Piter Z. Garcia Bautista

Graduate Assistant in Quantum & AI Research | Applied AI, Data Science, Reproducible Systems

📍 Rochester, NY 📞 +1 (646) 288-3300 ✉️ pzg8794@rit.edu
🌐 linkedin.com/in/garciapiterz 🐙 github.com/pzg8794 🌐 garciapiterz.com
ORCID: 0009-0008-5129-2820 ResearchGate profile

Summary

Graduate researcher with faculty-facing experience in quantum and AI research, applied data science, healthcare analytics, and reproducible evaluation. Current work focuses on threat-aware quantum routing, qubit allocation, and research infrastructure, while the broader research direction connects fairness, equity, and performance in quantum and clinical settings. Brings prior industry experience building Python- and pandas-based analytics workflows in healthcare and production environments, with strong experience in data cleaning, exploratory analysis, supervised machine learning, and reproducible reporting.

Technical Skills

Programming: Python, R, Java, C++, C#, SQL, MATLAB, JavaScript, Bash

Applied AI / ML: scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, supervised learning, feature engineering, model evaluation, fairness-aware ML

Data Science: pandas, NumPy, SciPy, Matplotlib, Plotly, Seaborn, data cleaning, EDA, statistical analysis, benchmarking

Cloud / Reproducibility: AWS (*primary industry cloud*), GCP (*research workflows*), Docker, Git/GitHub, Linux/Unix

Computational Research: bioinformatics pipelines, RNA structure modeling, reproducible experiments, technical reporting

Relevant Experience

Graduate Assistant – Quantum & AI Research

Fall 2025 – Present

Software Engineering Department, Rochester Institute of Technology

Rochester, NY

- Built a Python-based **multi-testbed experimental framework** to support **threat-aware quantum routing**, online path selection, and qubit allocation using **multi-armed bandit (MAB)**, contextual bandit, and neural decision methods.
- Build and maintain comparison-ready evaluation workflows across stochastic and adversarial threat environments for benchmark quantum-network routing and allocation experiments.
- Develop rerunnable workflows across **Colab, local, and GCP** environments with a shared research hub built through Colab, GCP-backed pipelines, automated validation, structured logging, shared datasets, and reproducible experiment tracking.
- Evaluate AI-driven routing methods through large-scale experiments, generating benchmark analyses, technical reports, and manuscript-ready research artifacts for faculty-led publications and follow-on research.

Teaching Placement – Computer Science and Inclusion

2025 – Present

Pine Brook Elementary School

Rochester, NY

- Support elementary computer science instruction in Pine Brook's IGNITE program through observation, assistantship, co-teaching, and gradually increasing lesson leadership.
- Co-design and deliver standards-aligned lessons integrating coding, robotics, digital fluency, and responsible AI concepts through hands-on, project-based learning for elementary students.
- Use differentiated materials, formative checks, and UDL-informed supports to strengthen accessibility and participation, while translating complex technical ideas into clear, usable workflows for students and collaborators.

Lead Researcher – EQUITAS and the Puzzle Plan

2024 – Present

Independent Research Initiative, RIT & University of Rochester

Rochester, NY

- Develop a cross-institutional research program linking healthcare analytics, RNA bioinformatics, fairness-aware diagnostics, bandit-based decision systems, disability-informed governance, and inclusive education.
- Build the technical backbone of this work through hospital readmission prediction, RNA secondary structure modeling, fairness-aware bioinformatics, and diagnostic AI governance, always linking predictive performance to equitable outcomes.
- Operationalize fairness, equity, and performance as joint design goals through fairness audits, mitigation strategies, governance narratives, workshop design, and community-informed research documentation.

Data Solutions Engineer

Sep 2022 – Aug 2024

VEDADATA Inc.

New York, NY

- Designed and implemented **Python-based data pipelines** and analytics workflows for business applications, emphasizing large-scale ingestion, cleaning, preprocessing, and validation.
- Built maintainable cloud workflows on **AWS**, supporting reproducible data transformation, and downstream analytics readiness.
- Applied quality checks, statistical analysis, and practices to improve traceability, reliability, and usability of production data systems.

AI Data Solutions Engineer

Jan 2021 – Sep 2022

VIOME Inc.

New York, NY

- Developed **applied AI and machine learning workflows** in Python for healthcare use cases, including preprocessing, feature engineering, model experimentation, and evaluation.
- Implemented supervised ML workflows using **scikit-learn** and related tools to support predictive and recommendation-oriented health applications.
- Worked extensively in **AWS**-based environments to support reproducible model development, deployment-oriented data preparation, and reliable evaluation pipelines.

Education

Master of Science in Data Science

Expected Aug 2026

Rochester Institute of Technology

Rochester, NY

- GPA: 3.9+ | Research focus: Fairness-Aware AI in Quantum & Clinical Settings
- Relevant Coursework: Data Science Foundations, Software Engineering for Data Science, Bioinformatics Algorithms, Data-Driven Knowledge Discovery, Neural Networks, Applied Statistics

Master of Science in Teaching Computer Science K–12

Expected Aug 2026

University of Rochester, Warner School of Education

Rochester, NY

- GPA: 3.9+ | NSF Robert Noyce Scholar | Advanced Certificate in Leadership in Disability and Inclusive Practices
- Relevant Coursework: Disability and Schools, Assessment in Instructional Contexts, Inclusive Classrooms, CS Standards Integration

Master of Science in Computer Science

2015

Rochester Institute of Technology

Rochester, NY

- Concentrations: Artificial Intelligence, Big Data Analytics, Computer Graphics

Bachelor of Science in Computer Engineering

2012

Farmingdale State College

Farmingdale, NY

- Dual Degree: Electrical Engineering & Computer Engineering Technology

Relevant Awards & Recognition

- **NSF Robert Noyce Scholar** (2024–2026)
- **MS International Scholarship** (2012–2015) – merit-based graduate funding through RIT and MESCYT
- **Honorable Mention Technology Award** (2012) – NYS STEP/CSTEP capstone recognition
- **IEEE Alumni Member** (2009–Present)

Research Advisers and Collaborators

Active research advisers and collaborators connected to my work.

- **Dr. Daniel Krutz and Dr. Travis Desell** – Faculty collaboration and supervision in the current quantum and AI graduate assistantship work, including experiment infrastructure, validation workflows, paper development, quantum routing, and multi-testbed research systems
- **Dr. Qi Yu** – Guidance in fairness-aware machine learning, data-driven knowledge discovery, and the development of equitable bioinformatics work
- **Dr. Fernando Rodriguez** – Bioinformatics and computational biology guidance for RNA secondary structure modeling, biomarker analysis, and diagnostic decision-support research

Formal references, portfolio materials, code samples, and research papers are available upon request.